

ABHAY KUMAR

AI/ML Engineer | Data Science | Python Developer

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PROFESSIONAL SUMMARY

AI/ML engineer building end-to-end intelligent systems — from data pipelines through model development to production deployment. Project experience spans graph-based AML/fraud intelligence, graph neural networks, AIOps reliability platforms, and distributed backend infrastructure (Kafka, Docker, FastAPI, RAG). Currently completing a market-data and document-intelligence pipeline aimed at quantitative finance AI engineering roles. Seeking a data science / machine learning engineering internship.

EDUCATION

Bachelor of Engineering (B.E.), Artificial Intelligence and Machine Learning

Sri Krishna Institute of Technology, Bengaluru — Visvesvaraya Technological University (VTU) | 7th Semester

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, Polars, PostgreSQL, DuckDB

Machine Learning: scikit-learn, LightGBM, Feature Engineering, Model Evaluation, Statistics, Deep Learning

AI Engineering: LLMs, RAG, LangChain, FAISS, Agents, Prompt Engineering, Graph ML

Backend & Systems: FastAPI, REST APIs, Redis, Docker, Kafka, Distributed Systems, OpenTelemetry

Tools: Git, GitHub, Jupyter Notebook, VS Code

PROJECTS

GraphShield AML — *Graph-Native AML Intelligence Platform*

- Built a graph-based anti-money-laundering investigation platform combining network analysis and temporal feature engineering with machine learning to surface suspicious transaction patterns.
- Added a policy-retrieval (RAG) layer to ground investigation recommendations in compliance policy documents, with built-in explainability for every flagged case.

Stack: Graph ML, RAG, Polars, DuckDB, FastAPI

SentinelOps AI — *Reliability Intelligence & Failure-Prevention Platform*

- Developed an AIOps platform combining telemetry ingestion, failure prediction, and causal root-cause analysis (RCA) to detect and prevent system failures before they escalate.
- Implemented safety-gated automated remediation with verified recovery, reducing manual incident-response steps.

Stack: Kafka, OpenTelemetry, Docker, Causal RCA, SRE tooling

Mercury X — *AI Workload Orchestration*

- Built a scheduling and orchestration system for AI compute workloads, optimizing resource allocation across concurrent jobs.
- Designed for reliability under variable load using distributed-systems principles.

Atlas X — *AI Governance & Mission Control*

- Designed a governance and monitoring layer for AI agents, providing oversight and control over autonomous AI behavior in production-like settings.

GraphCite-GCN — *Citation Network Classification with Graph Convolutional Networks*

- Built a graph neural network to classify research papers by modeling the citation network as a graph (papers as nodes, citations as edges), so each paper's prediction incorporates context from its citation neighborhood rather than isolated features alone.
- Implemented a multi-layer GCN architecture in PyTorch Geometric, aggregating neighbor and neighbor-of-neighbor representations across stacked graph convolution layers.
- Trained and evaluated on the Cora citation network (2,708 papers, 10,556 citation edges, 1,433-dimensional features, 7 classes).

Stack: PyTorch Geometric, Graph Neural Networks (GCN), Python

SOFT SKILLS

Problem Solving | Analytical Thinking | Communication | Fast Learner | Team Collaboration | Time Management